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An integrated material-structural analysis of prestress concrete affected by corrosion of non-prestressed reinforcement

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Abstract

Corrosion-induced concrete cracking is a significant stage of structural deterioration in prestressed concrete (PC) structures. Most current research predominantly focuses on the corrosion of prestressed steel strands, while paying limited attention to the study of non-prestressed reinforcement corrosion. In this study, an integrated material-structural numerical approach was developed to simulate corroded PC beams, considering corrosion products migration in pores and cracks. The cracking patterns and prestress losses obtained from the simulation results agreed well with experimental observations. Subsequently, using the proposed numerical approach, the interaction mechanism of the prestress level and reinforcement corrosion was investigated. The simulation results indicated that corrosion of longitudinal reinforcement leads to cross-sectional damage, causing the redistribution of stress across the section and increasing long-term deformation, ultimately resulting in prestress losses. Meanwhile, stirrup corrosion leads to an initial increase in prestress due to expansion forces when the degree of corrosion is not so high, but finally it will also lead to a degradation of mechanical performance.

Keywords

corrosion damage, creep, finite element analysis, prestressed concrete beams, prestress losses

Introduction

Prestressed concrete structures have become a widely favored choice in modern construction due to their outstanding load-bearing capacity, extended service life, excellent crack resistance, and minimal deflection development (Han et al., 2015; Nawy, 1996; Zhou et al., 2023). During the initial adoption of prestressed concrete structures, engineers commonly believed that their durability was relatively high due to the restriction of crack propagation. Despite these advantages, the durability of prestressed concrete structures can be significantly compromised by factors such as reinforcement corrosion (Dai et al., 2020; Pape and Melchers, 2011; Schupack and Suarez, 1982; Wang et al., 2019a; Zhang et al., 2017) and frost damage (Cao et al., 2016; Qin et al., 2016). In the context of concrete structure durability, reinforcement corrosion has long been regarded as a key concern (Bažant, 1979; Nilsson, 2019). Corrosion can reduce steel cross section, induce concrete cracking, degrade bond strength, lead to prestress loss, and eventually deteriorate the capacity of PC structures (Biondini and Frangopol, 2016; Li et al., 2023; Val et al., 2009). The main cause of some recent failure cases of existing PC structures is considering to be corrosion (Lim et al., 2016; Wang et al., 2023). The corrosion-affect long-term performance should be thoroughly investigated to ensure the serviceability and safety of PC structures and promote sustainable development of society (Bao et al., 2022a; Fan et al., 2022; Lao et al., 2024; Mao et al., 2022; Tong et al., 2023; Xia et al., 2024; Xiang et al., 2023).

In recent years, many researchers have conducted experimental studies to investigate concrete cracking and performance degradation induced by corrosion. Corrosion

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can reduce prestressed strand cross section, induce concrete cracking, degrade bond strength, lead to prestress losses, and eventually deteriorate the capacity of PC structures (Dai et al., 2020; Elfergani et al., 2013; Li et al., 2011; Recupero and Spinella, 2019; Tu et al., 2019; Val et al., 2009; Wang et al., 2019a; Yu et al., 2022; Zhang et al., 2016). Wang et al. (2019a) investigated the prestressed strand corrosion-induced concrete cracking behaviors under various prestress level experimental and analytically. And Yu et al. (2022) found corrosion of prestressing tendons led to a deteriorated flexural bearing capacity of a beam. Furthermore, Zhang et al. (2016) presented the fatigue behavior of corroded prestressed concrete beams, demonstrating that the corroded beams exhibited faster stiffness degradation, increased deflections, and wider crack widths under cyclic loading. In addition, prestress loss is an inevitable issue that prestressed structures face during their service life. Dai et al. (2020) investigate the effective prestress in eight corroded pretensioned concrete beams under various stress levels, determining that prestress losses depend on the degree of corrosion, and effective prestress reduces to zero when corrosion loss reaches 34.0%.

By analyzing historical literature, we found previous research has mainly focused on the issue of prestressed strand corrosion in PC structures, with little attention given to the corrosion of non-prestressed reinforcement. In fact, compared to prestressed steel strands, non-prestressed reinforcement is more prone to corrosion problems under high erosion conditions due to fewer protective measures and smaller concrete cover thickness (Zhao and Jin, 2016). How does the corrosion behavior of non-prestressing steel impact the stress levels of prestress. And how does the compressive stress within concrete affect the expansion behavior of corrosion products? In view of the first problem, the research (Wu et al., 2019; Zhang et al., 2022) in the field of concrete structure can bring us some enlightenment. Wu et al. (2019) found that with the increase of compressive stress, the corrosion damage of longitudinal reinforcement increased, while the corrosion damage of stirrup decreased. However, there are few related studies on the second question. Fundamentally, these two questions are interrelated for PC structures, but current research has yet to provide a clear explanation.

The establishment of the prediction model is another important issue for investigating corrosion-affect mechanical performance. The most widely applied corrosion-induced cracking model is the two-dimensional thick-walled cylinder model proposed by Bažant (1979). Subsequently, many studies have modified this thick-walled cylinder model, considering factors such as the filling capacity of corrosion products (Liu, 1996; Zhao et al., 2014, 2016) and mechanical properties of corrosion products (Lu et al., 2011; Ouglova et al., 2006). These models are more accurate when applied to just corrosion-damaged specimens, but they face challenges when applied to in-service structures under coupled external loading and environmental conditions. As a supplementary analytical tool, numerical simulations provide a valuable method to investigate the behavior of non-prestressed reinforcement corrosion in prestressed structures (Allaire, 2007; Bao et al., 2022b; Wang et al., 2019b, 2021; Wang and Maekawa, 2022; Yu et al., 2024). Numerical models can provide insights into the long-term performance of concrete structures under various environmental and loading conditions (Gebreyouhannes and Maekawa, 2016; Hansen and Saouma, 1999; Li et al., 2014; Michel et al., 2016; Ren et al., 2015; Wang et al., 2022), enabling researchers and engineers to formulate effective corrosion prevention strategies.

In response to these challenges and knowledge gaps, this study adopts a comprehensive approach to investigate the influence of non-prestressed reinforcement corrosion in PC structures. By integrating experimental research and numerical simulations, this study aims to elucidate the effects of prestress on reinforcement corrosion behavior and its subsequent impact on effective prestress levels. Ultimately, the goal of this study is to enhance the long-term performance and sustainability of prestressed concrete structures in corrosive environments.

Calculation models

The computational model employed in this study is based on the *DuCOM-COM3* multiscale simulation platform, as depicted in Figure 1. Gebreyouhannes and Maekawa (2016) introduced a numerical model that considers the impact of reinforcement corrosion on the stress within reinforced concrete elements. *DuCOM-COM3* comprises a thermo-hydro system that factors in concrete mix proportions, ambient temperature and relative humidity to compute concrete compressive strength, porosity, drying shrinkage, and creep. Additionally, it encompasses a mechanical system for structural analysis. Once the thermo-hydro system determines the mechanical properties of concrete, these properties are used in the mechanical system for structural analysis. In particular, when concrete experiences cracking due to loading, creep, or drying shrinkage, the extent of damage is incorporated into the thermo-hydro system in the subsequent time step. This enables the simulation of accelerated concrete deterioration. Furthermore, prestress losses are updated in the following calculations to account for the influence of stress on corrosion. Consequently, a more realistic analysis of the long-term structural behavior of PC members is achieved by considering time-dependent properties of concrete, including cement hydration, cracking, drying shrinkage,

creep, and reinforcement corrosion, alongside prestress losses.

Expansion stress model for reinforcement corrosion

Based on the fact that corrosion products can migration into pores and cracks (Michel et al., 2014; Wang et al., 2022; Wong et al., 2010; Zhao et al., 2016), corrosion products are classified into solid and gels of liquid phase. And multi-phase theory (Biot, 1941, 1963; Maekawa et al., 2008) is applied to propose a numerical approach that can predict the corrosion expansion, corrosion product migration and crack propagation behaviors in RC elements, as shown in Figure 2. In this study, concrete adopts a dual sklt-pore unit (combination of solid skeleton and porous media, where water and corrosion gel can move within the pores and cracks). Based on the mechanics of porous media, we derive the mass conservation governing equation (Maekawa et al., 2008),

$$\rho = (1-n)\rho_c + n\rho_f \tag{1}$$

where, ρ is the total density of concrete with corrosion gel which could fill the whole pore volume, ρ_c is the density of concrete skeleton, ρ_f is the density of corrosion gel, and n is defined as the pore volume ratio. The motion of corrosion gel is defined by the relative displacement from the concrete skeleton. Then, the movement of corrosion gel denoted by U_i is defined by its displacement relative to the concrete skeleton denoted by u_i and the mass-substantial relative displacement of corrosion gel phase denoted by w_i as (Wang et al., 2023),

$$U_i = u_i + w_i/n \tag{2}$$

The original concept of the Biot's theory in geotechnical engineering assumes that the porous medium is isotropic, meaning that individual soil particles are surrounded by a liquid matrix. Similarly, for the concrete surrounded by

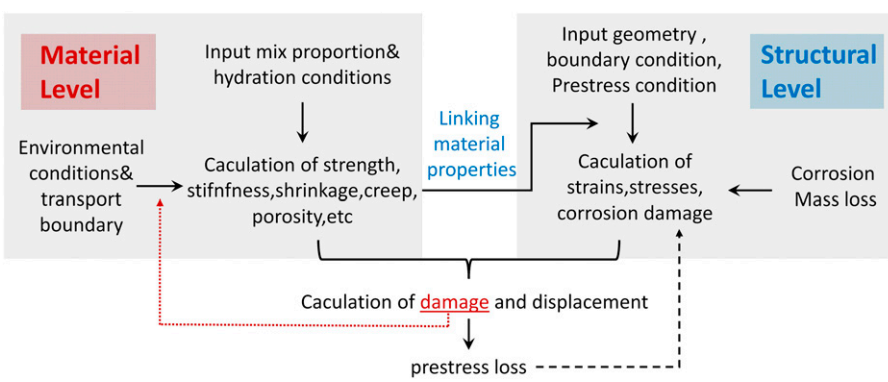

Figure 1. Framework of numerical simulation.

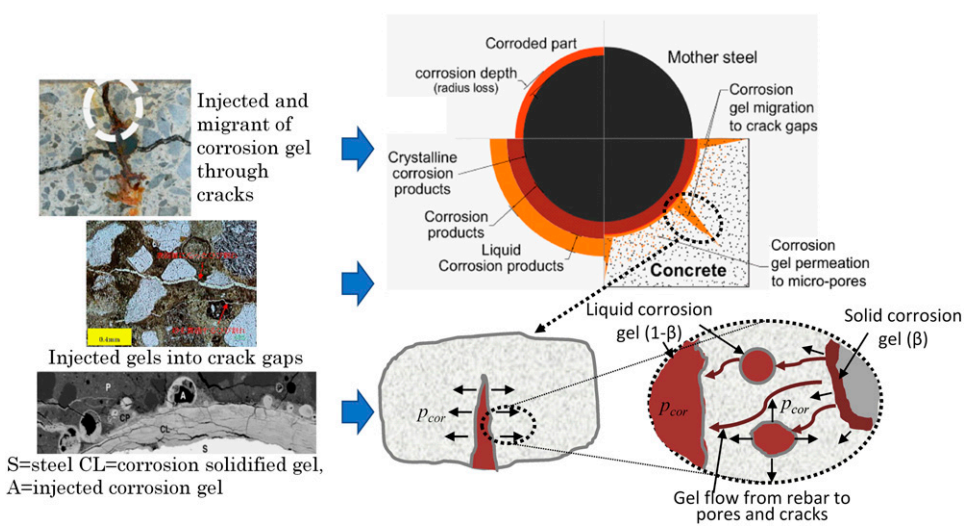

Figure 2. Reinforcement corrosion expansion-migration-induced cracking behavior.

corrosion gel, we express the total stress in the soil-liquid system (σ_{ij}) as a simple sum of the stress in the concrete skeleton and the corrosion gel's pressure term. However, the isotropic assumption of the porous medium is clearly not satisfied when the concrete cracks, thus we consider the anisotropic pore pressure brought about by the cracking of concrete as (Maekawa et al., 2003a),

$$\sigma_{ij} = \sigma_{ij}^{*} + \delta_{ij} p \tag{3}$$

$$\sigma_{ij} = \sigma_{ij}^{*} + \delta_{ij} \cdot l_i \cdot p \tag{4}$$

where p is defined as gel pressure, σ_{ij}* is the effective stress tensors defined on the solid skeleton, δ_{ij} is the Kronecker delta, and l_i is the unit directional vector normal to the crack plane. This isotropy assumption inside capillary pores of concrete may hold before cracking. When cracking occurs, an anisotropic gel pressure is generated. In formulating the dynamic equilibrium of the concrete skeleton and porous media in cracks, the interacting term between two phases is needed. Assuming that different kinematics of two phases may create mutual interacting forces, the governing equations can be formulated with the term of dragging forces rooted in the permeability (k_i) of corrosion gel phase in each i-direction through concrete pore and crack gaps as (Toongoenthong and Maekawa, 2005),

$$\sigma_{ij,j} + \rho g_i = \rho u_{i,mm} + \rho_f w_{i,mm} \tag{5}$$

$$p_{,i} = \rho_f (u_{i,mm} - g_i) + \rho_f w_{i,mm} / n + (1/k_i) w_{i,m} \tag{6}$$

where, the last term of equation (6) means the dragging action of pore gel and often referred as Darcy's law of motion, and subscript m and mm indicate the first and the second-degree differentiation with respect to time. In the case of soil mechanics, the skeleton constitutive model is applied for calculating σ_{ij}* as effective stress, and the constitutive model for soil particle assembly is used for simulation. The pressure of liquefied phase of corrosion gels is calculated as (Gebreyouhannes and Maekawa, 2016),

$$p = \overline{K}_f (w_{i,i} + \varepsilon_{ii} - (1-\beta) \cdot v_{cr}) < 0 \tag{7}$$

where, $w_{i,i}$+ε_{ii} represents the mean volumetric strain of gel phase inside concrete, v_{cr} is the net specific volume of the created gels inside a finite element, β is the ratio of the crystalline phase of gels. Then, the term (1-β) v_{cr} means the liquid corrosion gel's volume. The stresses caused by the solid part of gels are calculated with respect to σ_{ij}*. Through this formula, the mechanical interaction of cracked concrete and gel media is consistently installed in the framework of continuum mechanics. Assuming that the volumetric change of porous media is the simple summation of solid and pore gel due to Pascal's Principle of ignoring the shear stiffness of porous media, as shown in equation (8) (Hansen and Saouma, 1999).

$$\overline{K}_f = \left(\frac{1-n}{K_c} + \frac{n}{K_{cg}} \right)^{-1} \tag{8}$$

where, K_c and K_{cg} are bulk stiffness of concrete matrix and corrosion gel.

After the initiation of corrosion, some of the liquefied corrosion products permeates into the surrounding pores of concrete. In this study, the migration of un-crystalized gel to the capillary pores of concrete is considered by Washburn's equation (9) as (Gebreyouhannes and Maekawa, 2016),

$$V_{cpg} = V_p \left[\phi_{cp} \cdot e^{\left(-B_{cp} \cdot r\right)} \right], r = \frac{-2\sigma_{sf} \cos\theta}{p} \tag{9}$$

where, V_{cpg} is the amount of corrosion gel permeated to the capillary pores under the isotropic pressure denoted by p, V_p is the volume ratio of the cement paste matrix containing micro-pores, φ_{cp} is the total capillary porosity on V_p, B_{cp} is the distribution parameter corresponding to the capillary porosity and taken as $3.34*10^7$ (/m) (Maekawa et al., 2008), σ_{sf} and θ are the surface tension and the contact angle of the liquefied pore media [54] and σ_{sf} cosθ is assumed to be 0.1 in this study. The isotropic pressure denoted by p, induced by the increased volume of liquid corrosion gel and computed by equation (7), acts as the main force driving the liquid phase through the concrete micro-structure against the surface tension. The referential radius, denoted as r in equation (9), represents the limit size of the micro-pore that may contain the pore media. To simplify, a value of 0.4 is deliberately assigned to V_p, representing an elevated level, in order to accommodate the rise in local porosity due to calcium depletion resulting from corrosion products, as reported by Wong et al. (2010). The value of v_{cr} in equation (7) is calculated by subtracting V_{cpg} from the gross volume of created gels.

Based on the above introduction, we can calculate the stress state of concrete materials under the influence of corrosion expansion. By applying a multi-directional crack model (Koichi Maekawa and An, 2001) and constitutive laws (Maekawa et al., 2003a), we can calculate the corrosion damage and long-term deformations, as summarized in Figure 3. In the multiscale computational system, tri-linear curve is adopted for the constitutive law of reinforcing bar. In addition, the elastic-plastic and fracturing models are involved for compression creep analysis, where total strain is the sum of the elastic and plastic strain components and compressive stress is related to the elastic strain and compression fracture parameter (damage indicator) denoted by K_C as above. It means that the concrete nonlinearity in compression is represented by ε_p and K_C.

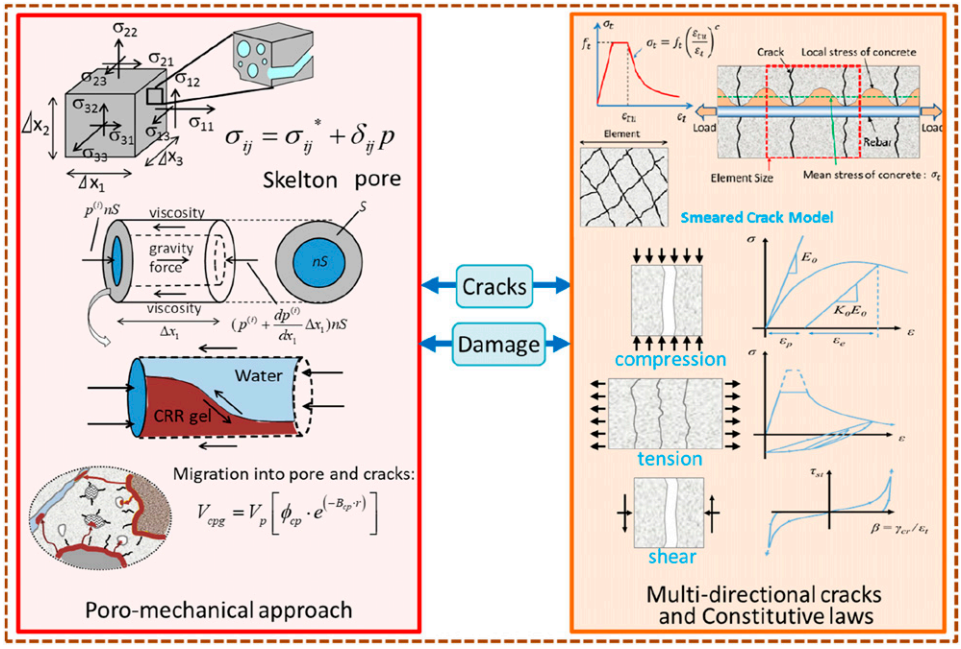

Figure 3. Calculation method for steel reinforcement corrosion damage and concrete crack.

The basic equations can be fully differentiated with respect to time and elasticity to form the-path-integral evolutional laws as equation (10) (El-Kashif and Maekawa, 2004; Maekawa et al., 2003a).

$$d\varepsilon_{\mathrm{p}} = \left(\frac{\partial \varepsilon_{\mathrm{p}}}{\partial t}\right)\mathrm{d}t + \left(\frac{\partial \varepsilon_{\mathrm{p}}}{\partial \varepsilon_{\mathrm{e}}}\right)\mathrm{d}\varepsilon_{\mathrm{e}}, dK_{\mathrm{C}} = \left(\frac{\partial K_{\mathrm{C}}}{\partial t}\right)\mathrm{d}t + \left(\frac{\partial K_{\mathrm{C}}}{\partial \varepsilon_{\mathrm{e}}}\right)\mathrm{d}\varepsilon_{\mathrm{e}} \tag{10}$$

The derivatives in terms of time increment indicate the time-dependent plasticity, and for plain concrete the term can be expressed as equation (11) (El-Kashif and Maekawa, 2004; Maekawa et al., 2003a).

$$\begin{aligned} \frac{\partial \varepsilon_{\mathrm{p}}}{\partial t} &= \phi\left(\frac{\partial \varepsilon_{\mathrm{p}}}{\partial t}\right)_{\mathrm{b}} + \kappa \\ \left(\frac{\partial \varepsilon_{\mathrm{p}}}{\partial t}\right)_{b} &= 0.034\left(\exp\left(\frac{\varepsilon_{\mathrm{e}}}{4}\right) - 1\right) \\ \kappa &= -\frac{1}{C_v}\left(\varepsilon_{\mathrm{p}} - C_{\mathrm{lim}}\varepsilon_{\mathrm{e}}\right), \kappa < 0 \end{aligned} \tag{11}$$

where, ϕ represents the nonlinear term, which is crucial at high level stress, and κ is a simply linear creep rate which can be derived from the creep model considering environmental condition and concrete pore structures at material level (Asamoto et al., 2006). In an RC element post-cracking, tension stiffening is manifested by accounting for the transfer of tensile stress through bond between reinforcing bars and concrete, which can be mathematically represented as below (Maekawa et al., 2003b).

$$\sigma_t = f_t(\varepsilon_{tu}/\varepsilon_t)^c \tag{12}$$

where σ_t and ε_t are the tensile stress and strain normal to crack plane, respectively; f_t is the tensile strength of concrete; ε_{tu} is the cracking strain; c is the stiffening coefficient, which is generally taken to be 0.4 for RC elements. The shear stress transfer in an RC element after cracking is modeled based on the contact density theory (Maekawa et al., 2006). More detailed information on the constitutive models can be found elsewhere (Maekawa et al., 2003b).

After obtaining the long-term deformation of concrete, it is easy to calculate the prestressing strand stress value and prestress loss through the deformation coordination condition. The calculation flow chart of prestress loss is shown in Figure 4.

Property parameters of corrosion products

Within the context of mechanical responses induced by corrosion expansion, the mechanical properties of corrosion colloids play a crucial role. These properties include the volume expansion ratio of corrosion products, the elastic modulus of corrosion products, the migration ratio of corrosion products, and the migration difficulty coefficient. These key parameters are crucial in the modeling of

the corrosion products expansion-migration-cracking behaviors, as shown in Figure 5.

Given the importance of the mechanical properties of corrosion products in modeling, the selection of corrosion product properties is based on prior experimental research and analysis. Regarding the expansion rate of corrosion products, the values of 5.0 were adopted, based on the findings under conditions of chloride exposure and electrochemical corrosion acceleration of Marcotte and Hansson (2007) and Takaya et al. (2013). The stiffness of corrosion products, with a value of 2 GPa, was referenced from Pease's research (Pease et al., 2012). As for the solid-liquid ratio of corrosion products and the permeability coefficient of liquid corrosion products, it's regrettable that no direct experimental data were available. Therefore, the values proposed by Gebreyouhannes and Maekawa (2016) were retained. In their study, they established a relationship between the solid-liquid ratio of corrosion products and the corrosion ratio. According to their reported values, the coefficient for the solid-liquid ratio that should be chosen for the corrosion rate is nearly 0.5 used in this study. Figure 6 indicates a comparison between the corrosion product solid-liquid ratio used in this study and the analysis of Gebreyouhannes and Maekawa (2016). The key corrosion product parameters used in this study are summarized in Table 1.

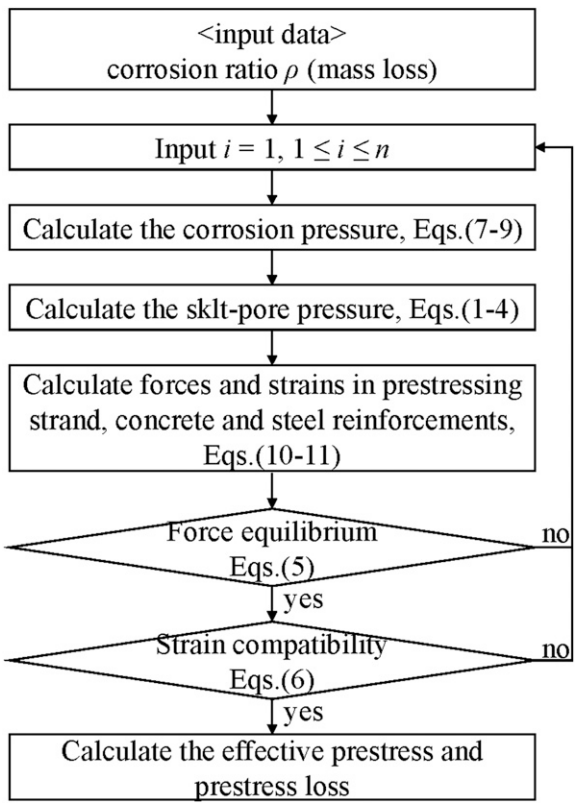

Figure 4. Calculation flowchart of prestress loss.

Experiments and model verification

Experimental setup

A total of four post-tensioned prestressed beams were fabricated in this test, with a cross-section size of 200 mm × 200 mm and a concrete length of 1.60 m. Including the anchoring system at both ends, the total length was 1.88 m. The longitudinal reinforcing bars of the prestressed beams had a diameter of 12 mm, while the stirrups were of 8 mm diameter, spaced at 200 mm intervals. The geometric dimensions and reinforcement layout of the prestressed beams are shown in Figure 7. The prestressed beams were made by using ready-mixed concrete with a strength grade of C40, a water to cement ratio of 0.5, and the mix proportion is shown in Table 2. The concrete's mechanical properties were measured based on GB50081-2019, with a 28-day compressive strength of 44.2 MPa and an elastic

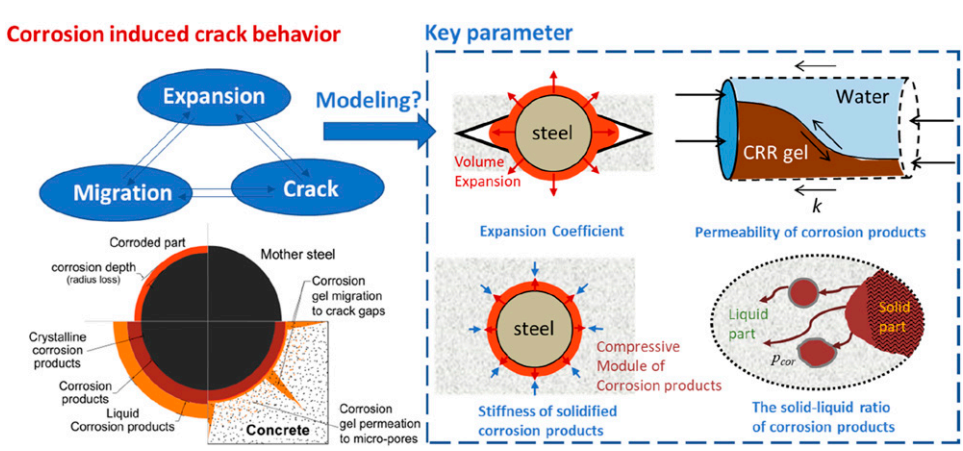

Figure 5. Key parameters in the corrosion products expansion-migration-cracking behaviors modeling.

modulus of 31 GPa. Three types of steel bars were used in the test: ordinary deformed steel bars, epoxy-coated steel bars (used for anti-corrosion), and steel strand (used for prestressing). The mechanical properties of the steel were provided by the manufacturer, as shown in Table 3.

After 43 days of wet curing for the concrete (with surface covering and regular watering), the prestressing of steel strand wires commenced. The initial stress applied to the prestressing steel strands was controlled at 70% of their tensile strength, which amounts to 1308 MPa. To replicate the degradation process observed in actual engineering structures as accurately as possible and to eliminate early prestress losses' interference with the test results (such as anchor slip, etc.), a wet-dry cycle followed by accelerated corrosion through electrical current was conducted after a natural curing period of 120 days. Four groups of specimens were employed, namely, None corrosion (*N*), Longitudinal reinforcement corrosion (L), Stirrup corrosion (S), and simultaneous corrosion of longitudinal reinforcement and stirrups (LS). None corrosion group (*N*) was used to isolate the effects of wet-dry cycle on internal concrete moisture and long-term deformation due to shrinkage. We take the specimen L as an example to illustrate the accelerated corrosion method used for the test. The longitudinal reinforcement of specimen L was made of ordinary steel, while the stirrup was made of epoxy-coated steel to prevent corrosion. The electrification process was carried out in parallel with the wet-dry cycling process, with the power supply being switched on during the wet phase and disconnected during the dry phase. The positive terminal of the DC power supply was connected to the longitudinal bars and the negative terminal was connected to the stainless steel mesh on

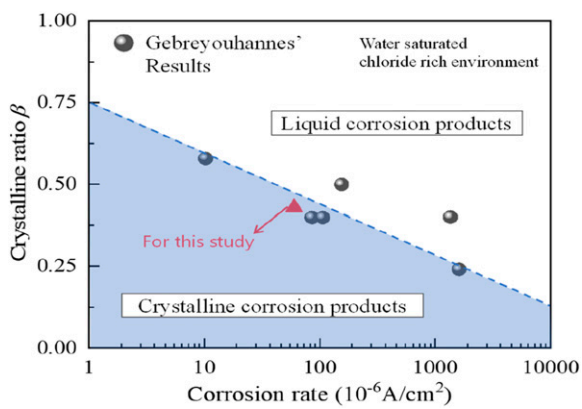

Figure 6. Comparison with solid-liquid ratios from existing analysis results.

Table 1. Key corrosion product parameters used in this study.

	Physical significance of parameter	Parameter	Value
Corrosion products	Stiffness of corrosion products	E_r	2 GPa (Pease et al., 2012)
	Expansion factors of corrosion products	a	5.0 (Marcotte and Hansson, 2007; Takaya et al., 2013)
	Permeability of corrosion products	k_r	5e-10 m/s^2 (Gebreyouhannes and Maekawa, 2016)
	The solid-liquid ratio of corrosion products	β	0.5 (Gebreyouhannes and Maekawa, 2016)

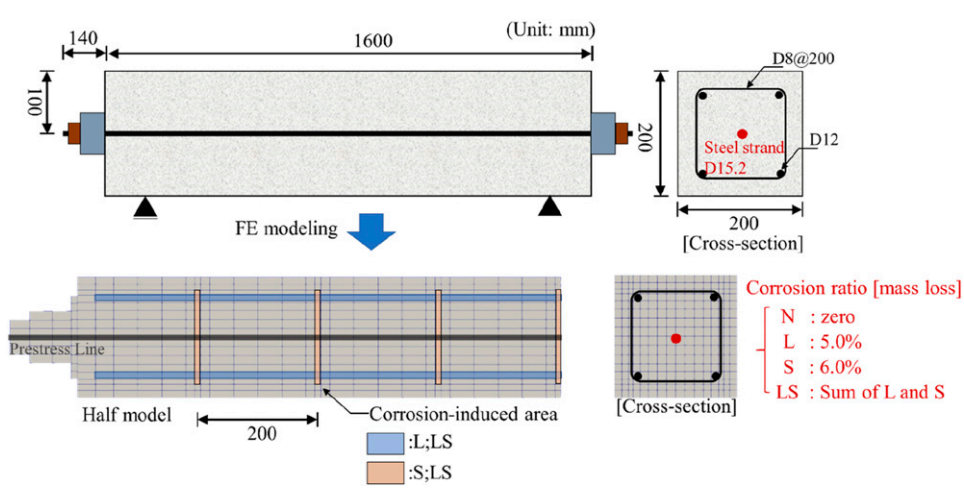

Figure 7. Dimensions and Details of RC beams and 3D FEM mesh.

Table 2. Materials information in the present study.

	Water (kg/m^3)	Cement (kg/m^3)	Coarse aggregate (kg/m^3)	Fine aggregate (kg/m^3)	Compressive strength (MPa)	Elastic modulus (GPa)
Concrete	185	370	1010	840	44.2	31
Steel	Type	Diameter (mm)	Yield stress (MPa)	Ultimate strength (MPa)	Elastic modulus (GPa)	
	HRB400 (ordinary)	8	435	606	205	
	HRB400 (ordinary)	12	435	606	205	
	HRB400 (epoxy-coated)	12	425	590	203	
	Strand	15.2	-	1908	196	

Table 3. Setting for reinforcement corrosion analysis at different Pre-Stress levels.

Sample	Reinforcement of corrosion	Pre-stress levels	Prestress after tension (MPa)	Prestress at initiation of corrosion (MPa)
L-F0	Longitudinal	Zero	Zero	Zero
L-F23	Longitudinal	0.23f_c	764.4	746.5
L-F40	Longitudinal	0.40f_c	764.4	713.6
L-F50	Longitudinal	0.50f_c	764.4	695.5
S-F0	Stirrups	Zero	Zero	Zero
S-F23	Stirrups	0.23f_c	764.4	746.5
S-F40	Stirrups	0.40f_c	764.4	713.6
S-F50	Stirrups	0.50f_c	764.4	695.5
LS-F0	Longitudinal and stirrups	Zero	Zero	Zero
LS-F23	Longitudinal and stirrups	0.23f_c	764.4	746.5
LS-F40	Longitudinal and stirrups	0.40f_c	764.4	713.6
LS-F50	Longitudinal and stirrups	0.50f_c	764.4	695.5

Note: 'L', 'S' and 'LS' represent the corresponding reinforcement of corrosion; 'FXX' represents a prestress level of XX% of the compressive strength of concrete (f_c).

the upper side of the concrete. Among these, the steel mass loss in the group with stirrup corrosion was 5.0%, while the groups with longitudinal reinforcement corrosion and both longitudinal and stirrup corrosion had steel mass loss of 6.0%. The test procedure is shown in Figure 8.

To specifically determine the impact of reinforcement corrosion, all subsequent results analysis is based on the difference between the group with corrosion (L,S and LS) and the group without corrosion (*N*). The data processing method for prestressing results involves subtracting the prestress change value of the corrosion group denoted by σ_X from that of the wet-dry cycling group of σ_N, thereby obtaining the isolated effect $\Delta\sigma_{CR\text{-}X}$ of corrosion, as shown in equation (12). It should be noted that for the discussion of prestressing results, data treatment used for the numerical analysis is consistent with the test results.

$$\Delta\sigma_{CR-X} = \sigma_X - \sigma_N$$

where, X represents the corrosion reinforcement, corresponding to longitudinal reinforcement corrosion (L), stirrup corrosion (S), and the simultaneous corrosion of longitudinal and stirrup reinforcement (LS).

Longitudinal reinforcement corrosion

Based on the experimental setup, an analysis of the corrosion-induced cracking behavior and long-term performance of PC beams affected by longitudinal reinforcement corrosion was conducted. The corrosion ratio of longitudinal reinforcement during the simulation process is considered to increase linearly with time. Figure 9(a) illustrates the development process of longitudinal reinforcement corrosion damage. As the reinforcement corrosion ratio increases, corrosion products progressively fill the voids, generating expansive stresses, resulting in cracks along the axial direction of the longitudinal reinforcements. These cracks align with the arrangement of the longitudinal reinforcements, consistent with the observed crack patterns in the experiments and in previous study (Wu et al., 2019; Zhu et al., 2023), as shown in Figure 9(d). In addition, it can be observed that the cracks on the

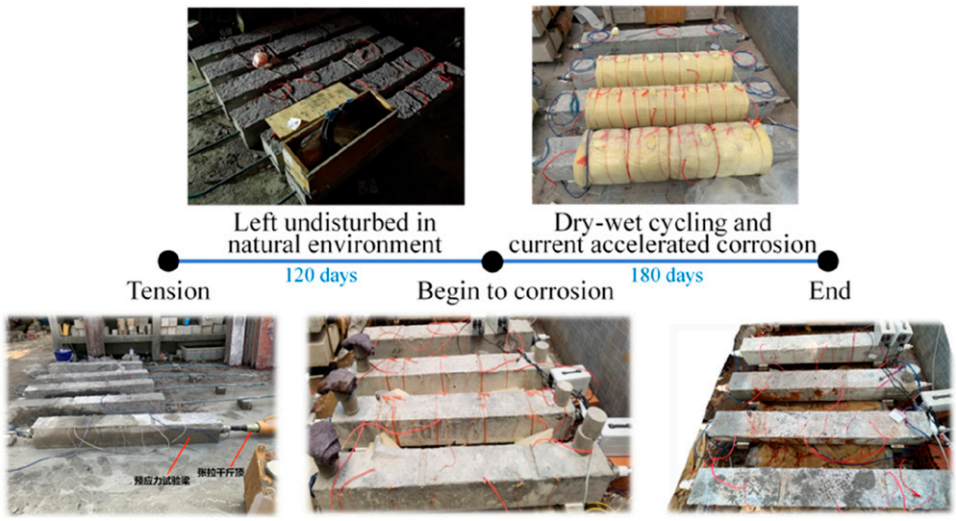

Figure 8. Test procedure in this study.

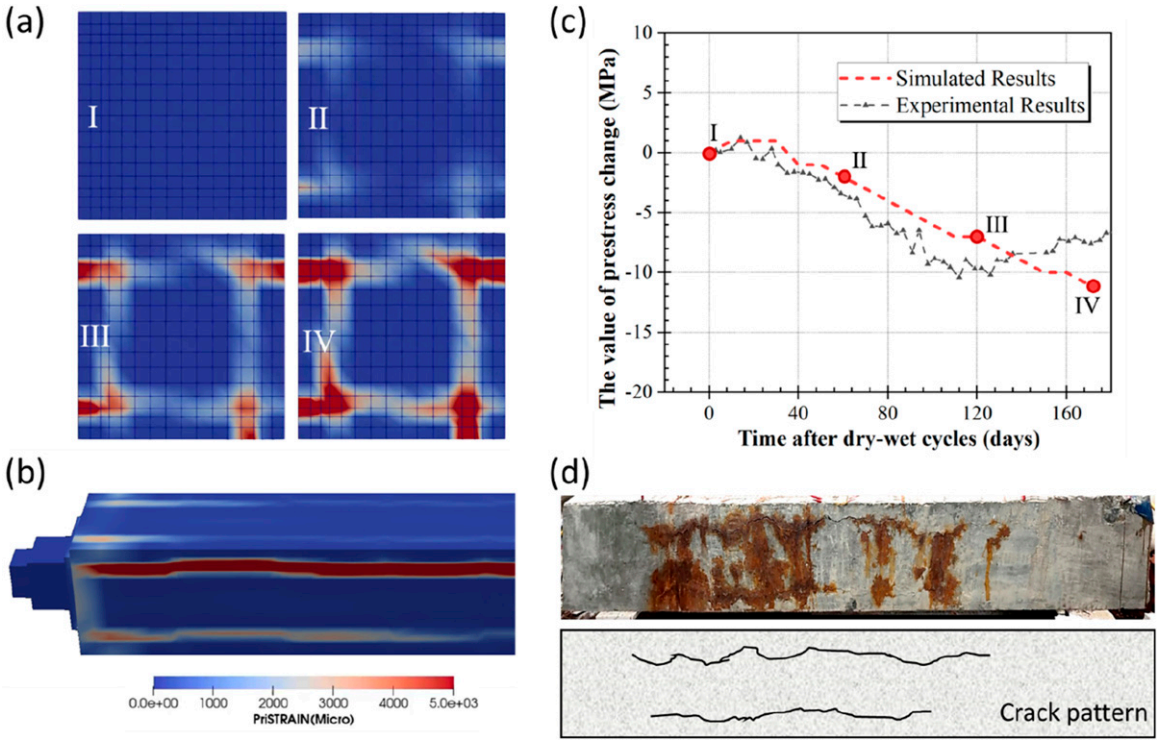

Figure 9. Experimental and simulation validation of longitudinal reinforcement corrosion. (a) development process of longitudinal reinforcement corrosion damage; (b) simulation results of concrete surface strain distribution; (c) the value of prestress change; (d) corrosion-induced concrete cracks in the experiment.

concrete surface do not propagate uniformly. This non-uniformity arises because, as the first microcracks form, subsequent corrosion products continuously fill these cracks, causing further expansion (See Figure 9(a)). Consequently, localized damage becomes more severe, while other potential cracking paths experience reduced expansion.

Figure 9(c) illustrates the relationship between prestress changes ($\Delta\sigma_{CR\text{-}L}$) in the prestressed beams and the corrosion process of the longitudinal reinforcements. Both simulation and experimental results indicate that the corrosion of the longitudinal reinforcements leads to prestress losses. It should be admitted that there is a certain deviation between the simulation results and the

experimental results. As shown in Figure 9(c), after 120 days, the prestress remains unchanged or even slowly increases, which seems difficult to explain mechanically. However, in fact, the result shown in Figure 9(c) is the result of subtracting specimen L and specimen *N*. We consider that the subtle differences in stress changes caused by factors such as shrinkage and creep between the two specimens may result in the stress remaining unchanged or even slightly increasing in the test data after 120 days. Experimental and simulation results both show the appearance of cracks on the concrete surface, accompanied by a loss of stiffness across the entire concrete cross-section. Consequently, the overall stiffness of the cross-section decreases, resulting in greater axial compression deformation under prestress, leading to prestress losses.

Stirrup corrosion

Similarly, the stirrup corrosion behavior was analyzed. Figure 10(a) illustrates development process of stirrup corrosion damage. The expansion induced by the corrosion of the stirrups occurs along the direction of their placement, aligning with the direction of prestress application. It can be observed that the damage caused by the corrosion of stirrups is relatively minor compared to the corrosion of longitudinal reinforcements (see Figure 10(b)). And the experimental results show the similar conditions that a tiny crack were observed, as shown in Figure 10(d). In fact, the limiting effects of stress and constraints on the expansion of liquid products have also appeared in previous studies, such as the freezing (Gong et al., 2018; Gong and Maekawa, 2018) and ASR expansion (Takahashi et al., 2016). The occurrence of this phenomenon is due to the stress driving the migration of corrosion products along the direction perpendicular to the stress direction. Figure 10(c) presents the prestress change ($\Delta\sigma_{CR\text{-}S}$) with the corrosion of the stirrups. In the case of stirrup corrosion, the prestress exhibits a clear increasing trend. This is different from our previous understanding, as corrosion-induced cracking damage due to stirrup corrosion, whether it is internal micro-damage or surface cracking, should reduce the concrete performance and lead to a decrease in prestress. However, from the experimental results, combined with the analysis of the corrosion expansion force caused by stirrup corrosion, it can be seen that the stirrup corrosion products generate longitudinal expansion force, which is applied to the concrete, causing the concrete compressive strain to recover, thus increasing the prestress. This self-strengthening effect might appear somewhat unexpected, but may be applicable in case of the lower degrees of corrosion damage for reinforcement. Further discussion of this phenomenon will be conducted in the subsequent analysis in Section 4.

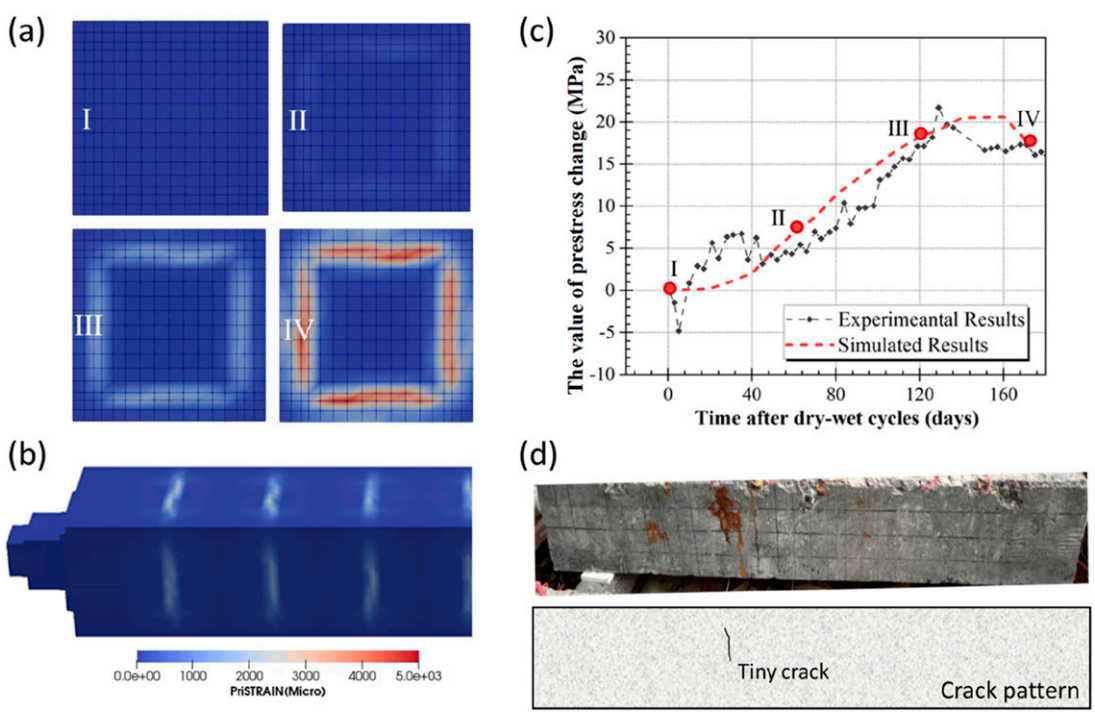

Figure 10. Experimental and numerical validation of Stirrup corrosion. (a) development process of stirrup corrosion damage; (b) simulation results of concrete surface strain distribution; (c) the value of prestress change; (d) corrosion-induced concrete cracks in the experiment.

Simultaneous corrosion of longitudinal reinforcements and stirrups

Figure 11(a) illustrates development process of simultaneous corrosion of longitudinal reinforcements and stirrups corrosion damage. It can be anticipated that the simultaneous corrosion of both the longitudinal reinforcements and stirrups would be a combination of their individual corrosion states. As previously mentioned, the corrosion of the longitudinal reinforcements exhibits severe cracking, while the surface of the stirrups shows no apparent damage. Accordingly, when both the longitudinal reinforcing bars and stirrups corrode simultaneously, the crack pattern resembles that of the corrosion of the longitudinal reinforcements, corresponding to the results observed in the experiment as shown in Figure 11(b) and (d). In the case of simultaneous corrosion of the longitudinal and stirrups, the prestress change during the corrosion process is relatively minor, as shown in Figure 11(c). However, the stress variation of Specimen LS is not simply the superposition of the stress variations of Specimen L and Specimen S. It can be observed that the prestress increase (20 MPa) corresponding to the corrosion of the stirrups is approximately twice as much as the value of the stress (10 MPa) decrease due to longitudinal reinforcement corrosion. Yet, in the case of simultaneous corrosion of the longitudinal and stirrups, the impact of both factors can be considered to cancel each other out. This is due to the fact that in the early stages of the corrosion of the longitudinal reinforcement, cracks are more likely to occur, and corrosion products extend along these cracks. Gradually, the damage to the concrete cross-section leads to a decrease in prestress. On the other hand, the mechanism of stirrup corrosion expansion is relatively smaller.

Interaction between prestress level and reinforcement corrosion

From the above discussion, the theory-based multiscale models, simulation results and the experimental measurements can be cross-verified with each other, which provide us more confidence for the further numerical investigation. The test specimens were constrained by the limitations of the tensioning equipment, resulting in prestress levels at both ends of approximately 23% of the ultimate compressive strength (f_c) of concrete. However, for common prestressed engineering applications, prestress levels typically fall within the range of 0.3 to 0.55 times the compressive strength of concrete. Furthermore, the stress state can affect the processes of reinforcement corrosion and expansion. To quantitatively analyze the impact of prestress levels on the corrosion process of non-prestressed reinforcement, four prestress conditions were established based on the experimental results. These conditions correspond to three different types of steel reinforcement corrosion. Furthermore, higher reinforcement corrosion ratios (up to 15%) were set to observe the changes in the

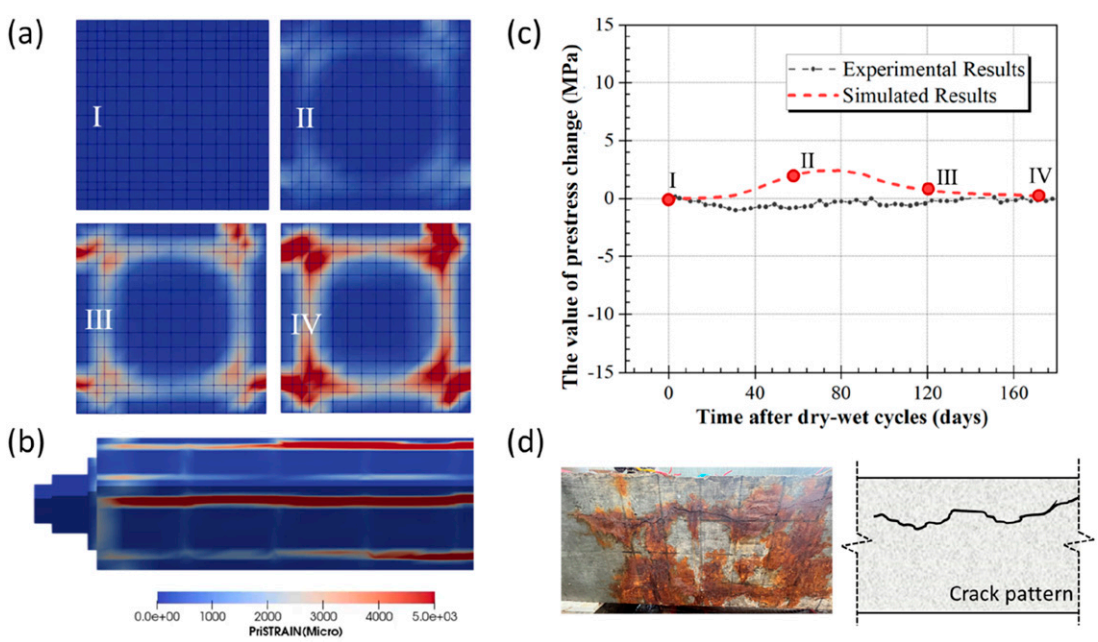

Figure 11. Experimental and numerical validation of simultaneous corrosion of longitudinal reinforcements and stirrups. (a) development process of reinforcement corrosion damage; (b) simulation Results of concrete surface strain distribution; (c) the value of prestress change; (d) corrosion-induced concrete cracks in the experiment.

performance of prestressed beams under more severe corrosion conditions. The setting for reinforcement corrosion analysis at different pre-stress levels is shown in the Table 3. It should be noted that the method used in the analysis for larger prestress values is to enlarge the area of the prestress line elements. Additionally, the stress values of prestressing steel wires at the moments of post-tensioning and the onset of corrosion are also presented in Table 3.

Longitudinal reinforcement corrosion

Figure 12 illustrates the distribution of principal strains on the surface and mid-span section of prestressed beams under different prestress levels. As the reinforcement corrosion ratio increases, corrosion products from the longitudinal reinforcements gradually expand, causing cracks in the concrete protective layer. Referring to L-F0 and L-F23, the influence of prestressing stress on the longitudinal reinforcement corrosion-induced expansion is relatively minor. Despite previous research indicating that concrete corrosion damage under load may be more significant (Wang et al., 2019a), it significantly differs from the stress conditions studied in this paper. When loads induce tensile stress in corroded steel-reinforced concrete, the combined external load stress and corrosion expansion forces may lead to protective layer cracking and further degradation. However, in the context of this study, prestress does not result in this phenomenon.

For more severe corrosion damage of longitudinal reinforcing bars, the changes in stress values of prestressed tendons under different prestressing levels are shown in Figure 13. This trend aligns with the experimental findings, as shown in Figure 9. The loss of prestress occurs due to corrosion-induced cracking and damage. As the concrete cross-sectional stiffness decreases, the concrete further compresses under tensile stress, leading to a continuous reduction in prestress. Furthermore, regarding the impact of

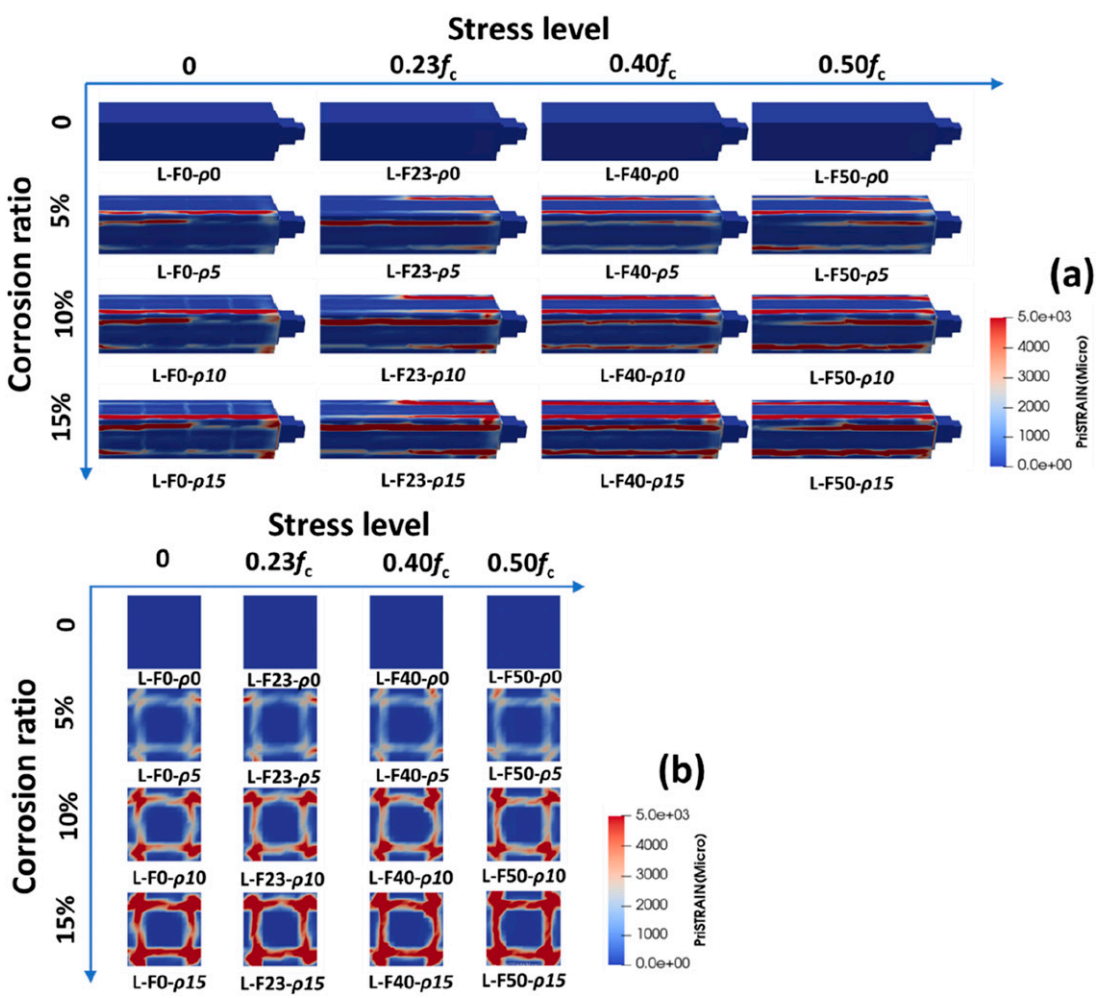

Figure 12. The influence of prestress level and corrosion ratio on longitudinal reinforcement corrosion. (a) surface of PC beams; (b) cross-sections of mid-span position.

prestressing stress levels, it is evident that higher prestressing stress levels result in more significant prestress losses. Importantly, this effect is not directly related to the degree of corrosion damage, as L-F23 and L-F50 exhibit nearly identical corrosion damage (see Figure 12). Instead, it occurs because, when the section experiences cracking damage, stress redistribution causes localized increases in concrete stress in undamaged areas. As we known, higher stress levels correspond to greater creep strain. Additionally, when comparing the prestress losses between L-F40 and L-F50, it's noticeable that L-F50 experiences more significant reduction. This might be attributed to the nonlinear nature of concrete creep behaviors becoming more apparent when the concrete stress level exceeds 0.4 times the compressive strength of concrete. As a result, concrete creep strain increases further, leading to more pronounced prestress losses. The mechanism by which longitudinal reinforcement corrosion affects prestress is illustrated in Figure 14. In summary, as the prestress level increases, the influence of longitudinal reinforcement corrosion on prestress losses in PC structures becomes more substantial.

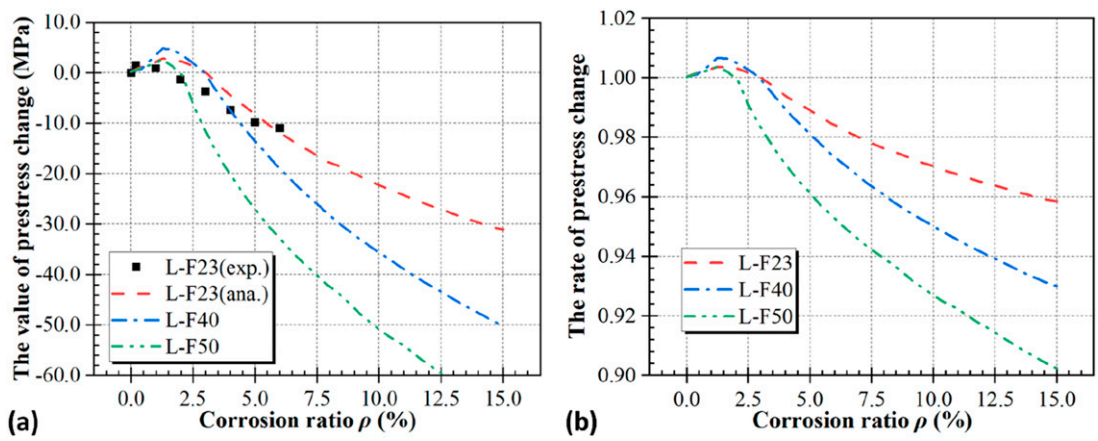

Figure 13. Effect of Longitudinal reinforcement corrosion rate on prestress losses under different prestress levels. (a)the value of prestress change; (b)the rate of prestress change.

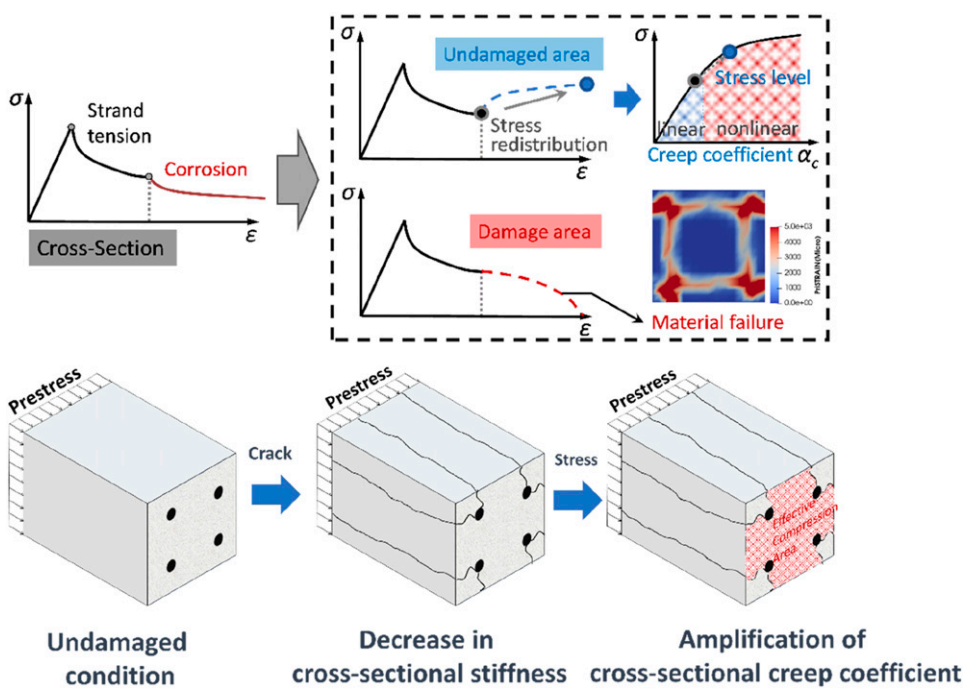

Figure 14. The mechanism that longitudinal reinforcement corrosion affects prestressing.

Stirrup corrosion

Figure 15 illustrates the damage of stirrup corrosion at different prestress levels. Comparing the simulation results for the cases with no prestress (S-F0) and a prestress level of $0.23f_c$ (S-F23), it can be observed that the corrosion damage to stirrups is significantly severe when no prestress is applied. As the expansion direction of the stirrups corresponds to the direction of prestress, and the presence of prestress limits the axial expansion of corrosion products and prevents concrete cracking. This phenomenon is similar to the expansion behavior of ASR (alkali-silica reaction) gel products in concrete pores under stress (Takahashi et al., 2016). Furthermore, comparing the extent of corrosion damage at different prestress levels, it is evident that as the prestress level increases, the corrosion damage to the stirrups gradually decreases, resulting in lighter surface damage to the PC beams.

The experimental results of stirrup corrosion indeed indicate that the corrosion of stirrups can be beneficial in maintaining the prestress of prestressed beams. However, this finding appears contradictory to the general understanding that corrosion leads to performance deterioration. Figure 16 illustrates the changes in prestress of the test specimens after more severe corrosion damage. With an increase in the reinforcement corrosion ratio, the change in prestress exhibits two stages. In the first stage, when the reinforcement corrosion ratio is limited, the degree of damage to the concrete section is minor (no significant damage to the surface of the concrete beam). Corrosion-induced expansion forces cause the concrete to expand along the direction of prestressed steel reinforcement, resulting in an increase in prestress. At this stage, the axial expansion force induced by stirrup corrosion predominates.

In the second stage, as the reinforcement corrosion ratio increases to a high level, the concrete surface experiences noticeable damage, corresponding to cracking of the concrete

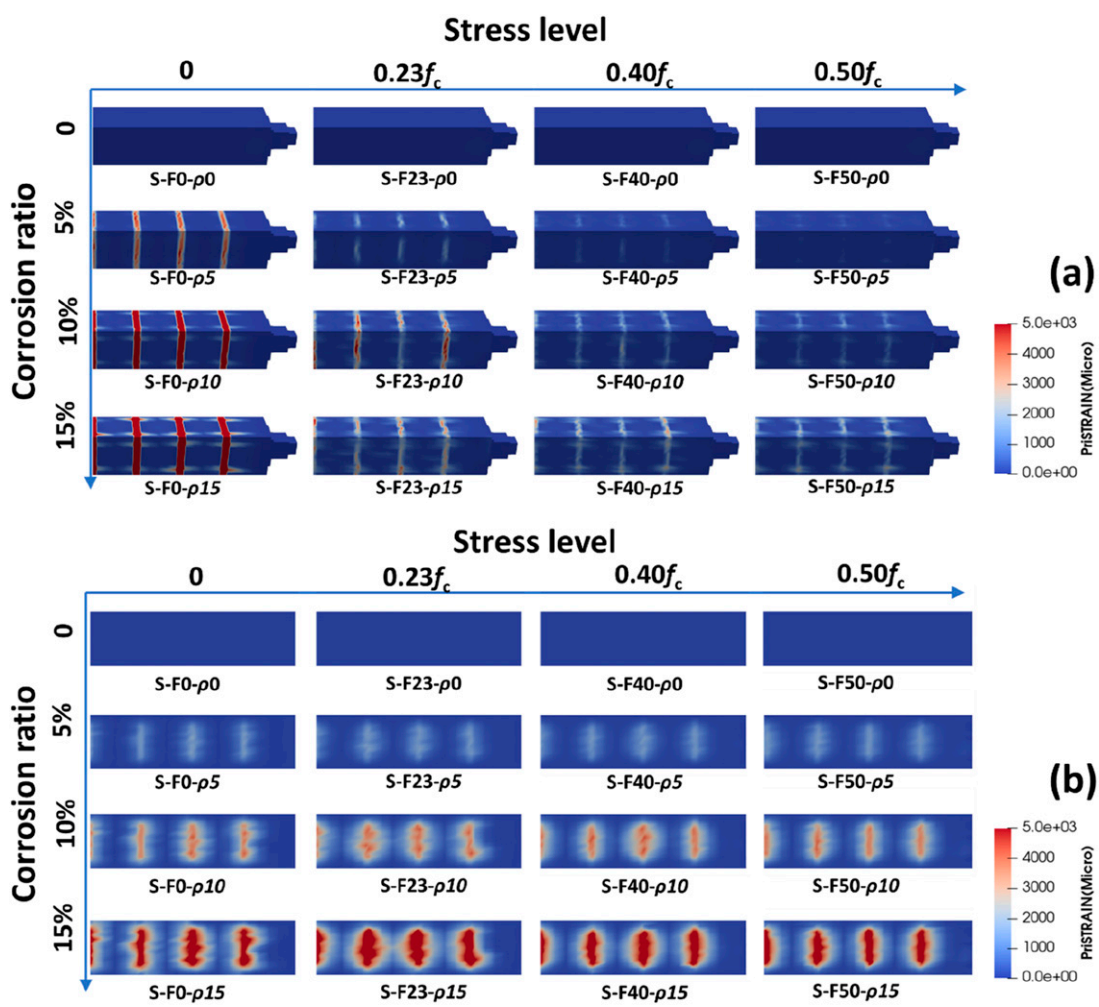

Figure 15. The influence of prestress level and corrosion ratio on stirrup corrosion. (a) surface of PC beams; (b) longitudinal section of stirrups position.

protective layer. During this stage, the expansion stress induced by corrosion almost stops increasing. However, as the reinforcement corrosion ratio continues to rise, the extent of damage to the concrete section gradually worsens, leading to a gradual decrease in prestress. Therefore, at higher reinforcement corrosion ratios, the prestress in the prestressed beam shows a decreasing trend. Referring to S-F23 and S-F40 in Figure 16, the influence of stirrup corrosion on prestressing loss in prestressed structures diminishes as the prestress level increase. This is also due to the restraining effect of prestress on the axial expansion of stirrup corrosion.

Simultaneous corrosion of longitudinal reinforcements and stirrups

Figure 17 depicts the development process of surface damage due to corrosion expansion with simultaneous corrosion of longitudinal reinforcements and stirrups (LS). The analysis results reveal that the situation when both longitudinal reinforcing bars and stirrups corrode simultaneously (LS) closely resembles the state of corrosion in only longitudinal reinforcement (L). This resemblance primarily arises from the faster development of corrosion in the longitudinal reinforcement, as demonstrated in Figure 9. The expansion of stirrup corrosion products may follow the path of longitudinal reinforcement damage. In other words, crack propagation can influence the damage pattern of corrosion products, akin to previously reported coupled damages of freeze expansion of concrete and ASR one (Gong et al., 2023). Consequently, when both longitudinal reinforcing bars and stirrups corrode simultaneously, the corrosion damage state aligns more closely with the condition of corrosion in only the longitudinal reinforcement.

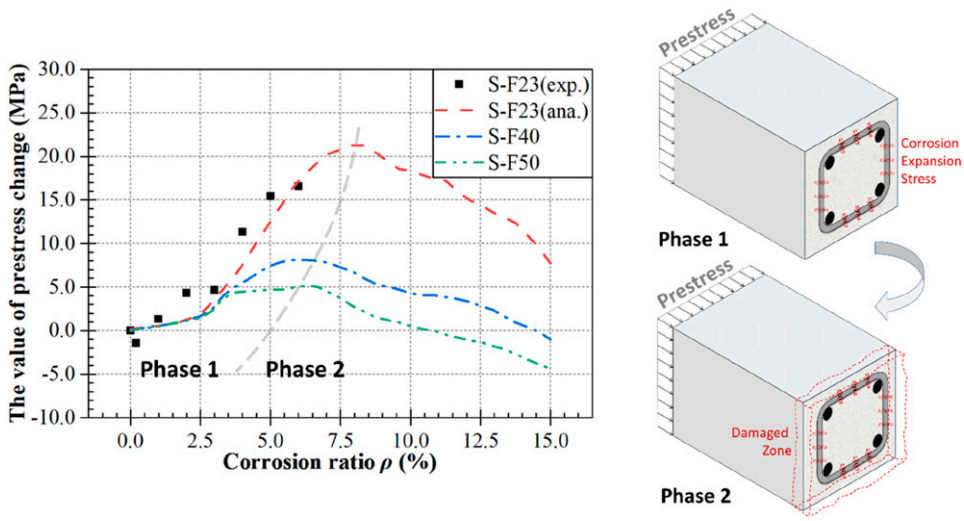

Figure 16. Effect of Stirrup corrosion rate on prestress losses under different prestress levels.

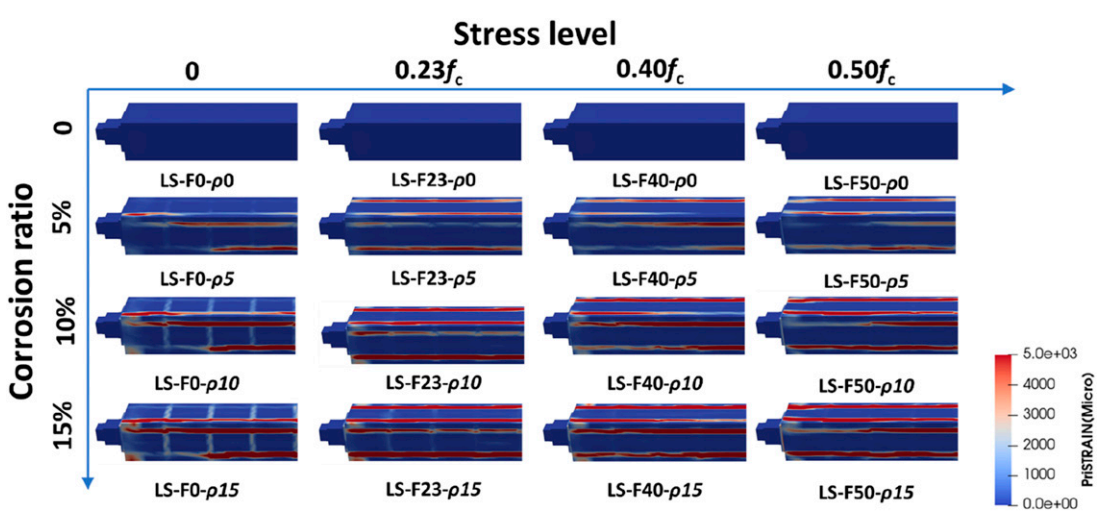

Figure 17. The influence of prestress level and corrosion ratio on surface corrosion damage of PC beams.

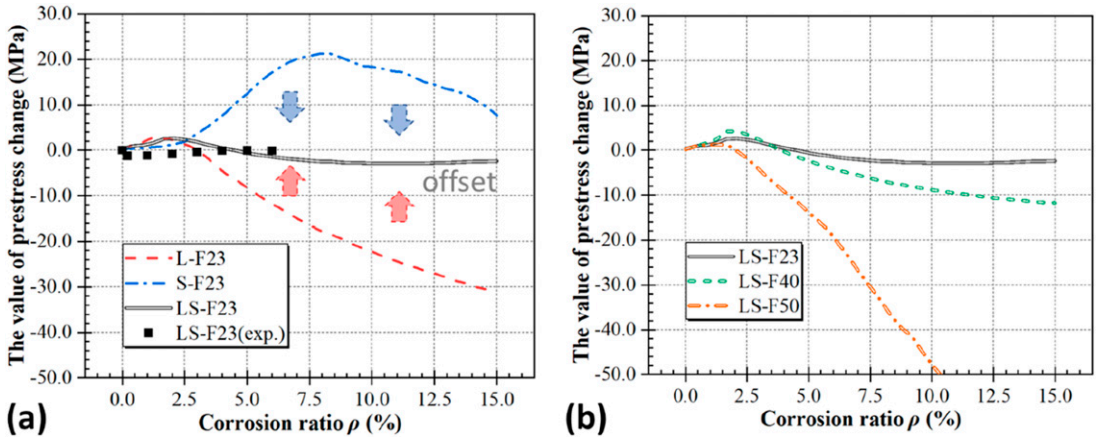

Figure 18. Effect of reinforcing bar corrosion ratio on prestress losses under different prestress levels. (a) the offset effect; (b) different prestress levels.

Correspondingly, because of the simultaneous corrosion of longitudinal reinforcement and stirrups, the changes in prestress fall between that of single-type reinforcement corrosion. Regarding the prestress level in the test specimens, as discussed in *Simultaneous corrosion of longitudinal reinforcements and stirrups*, at lower corrosion ratios, the influence of corrosion in longitudinal reinforcement and stirrup almost counteracts each other, as shown in Figure 18(a). However, with an increase in the degree of corrosion damage, the corrosion of both longitudinal and stirrup's reinforcing bars intensifies, causing cross-sectional damage and an increase in the overall axial deformation of concrete, which may lead to a decrease in prestress. Additionally, with an increase in prestress level, the prestress losses induced by the corrosion of longitudinal reinforcement becomes more pronounced, thus making the trend in prestress change more closely resemble the situation of longitudinal reinforcement corrosion, as shown in Figure 18(b).

Conclusion

In this study, the prestress losses and concrete damage pattern of prestressed reinforced concrete beams subjected to reinforcement corrosion was investigated. The detailed conclusions are earned as follows:

1) By quantifying the pore mechanical properties of corrosion products, the influence mechanism of non-prestressed longitudinal reinforcement corrosion and stirrup corrosion on the change of component prestress was revealed.
2) Corrosion of longitudinal reinforcement leads to a decrease in section stiffness and causing prestress losses. Additionally, stress redistribution leads to further creep development and subsequent prestress losses.
3) At low reinforcement corrosion degrees, the corrosion expansion forces generated by stirrups corrosion increase the prestress. However, when severe damage occurs in the concrete region near stirrups, stirrups corrosion causing a reduction in stiffness and gradual prestress losses.
4) The higher the stress level, the greater the prestress losses in PC beams due to longitudinal reinforcement corrosion. On the contrary, the impact of stirrup corrosion gradually diminishes. In addition, the impact of simultaneous longitudinal reinforcement and stirrup corrosion falls between the scenarios of single longitudinal reinforcement corrosion and stirrup corrosion.

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